

Methodology

Part 1 - Calibration results

Part 2 - Seeding simulation overview

Part 3 - Extinction and steady-state

Part 4 - Primary outcome: Incidence per 1,000 patient-days

Part 5 - Prevalence (capped at 100%)

Part 6 - Epidemic trajectories

Part 7 - Summary table

Part 9 - Export

ARCANE - Incidence Calibration and Seeding Analysis

Code ▾

Novel pathogen introduction on the French inter-hospital transfer network

MESuRS/Cnam - Task 4.3

September 2026

Methodology

Study population and network

The French inter-hospital transfer network was derived from 2024 PMSI (Programme de Medicalisation des Systemes d'Information) administrative discharge data. The network comprises **2,529 hospitals** (etablissements de soins) including acute care (MCO - Medecine, Chirurgie, Obstetrique), rehabilitation (SSR - Soins de Suite et de Readaptation), psychiatric (PSY), home hospitalisation (HAD - Hospitalisation a Domicile), and mixed (MCO/SSR) facilities. Daily transfer counts between each pair of facilities were derived from rolling 7-day sums, divided by 7 to give daily averages and rounded to the nearest integer (minimum 1). Total network beds: 374,545.

Patient flow model

The model assumes **constant full bed occupancy** throughout the simulation. Daily patient turnover at each hospital is determined by observed daily admission counts from the 2024 PMSI (NO_ADMISSION_DAILY_DIRECT_HBN dataset; 0hospital-days). Because occupancy is assumed constant, daily discharges equal daily admissions at each facility. The pathogen transmission status of discharging patients is determined by a **hypergeometric draw** - sampling without replacement from the current colonised (I) and susceptible (S) patients at each facility - preserving the colonisation proportion among those leaving. Of discharged patients, a facility-specific fraction is transferred to other hospitals (via the transfer probability matrix P); the remainder are admitted from the community. The number of patients admitted from the community equals total admissions minus transfers received. If incoming transfers exceed a facility's admission quota on a given day, excess transfers are scaled down proportionally (capacity enforcement). Community-admitted patients may be colonised at admission with probability π (π_{vec}).

SIS transmission model

Within each hospital h on each day t , susceptible patients (S) can acquire colonisation from infected/colonised patients (I) according to a **frequency-dependent stochastic SIS (Susceptible-Infected-Susceptible)** process:

$$P(\text{new colonisation} \mid S, I, N) = 1 - \exp(-\beta \times I/N)$$

where β (β_{vec}) is the daily within-hospital transmission probability per susceptible-infectious contact (units: per day), $N = S + I$ is total hospital occupancy, and $\text{new_inf} \sim \text{Binomial}(S, p_{inf})$. Biological decolonisation occurs at rate γ (γ_{vec}) = 1/387 per day, corresponding to a mean carriage duration of 387 days for ESBL-producing Enterobacterales (ESBL-E). No patient isolation ($\alpha = 0$) is assumed.

Incidence calibration

Incidence targets were derived from ECDC EARS-Net 2022 antibiotic resistance carriage prevalence benchmarks, converted to hospital-acquired incidence per 1,000 patient-days using:

$$\text{incidence} = (\gamma + 1/\text{LOS}) \times (P_{\text{total}} - \pi) \times 1,000$$

where LOS is the network-weighted mean length of stay (9.28 days from PMSI data) and P_{total} is the ECDC carriage prevalence for each resistance tier. Three tiers are defined using ECDC official category boundaries:

Incidence calibration targets derived from ECDC EARS-Net 2022. CRE = Carbapenem-Resistant Enterobacterales. VRE = Vancomycin-Resistant Enterococci. MRSA = Methicillin-Resistant Staphylococcus aureus. ESBL-E = Extended-Spectrum Beta-Lactamase-producing Enterobacterales. pd = patient-days.

Tier	ECDC category	ECDC carriage range	Incidence target (/1,000 pd)	Epidemiological examples
Low	low (>1-10%)	1-10%	0.99-10.92	CRE in France/Germany, VRE in Scandinavia, MRSA in low-burden Northern Europe
Moderate	moderate (>10-20%)	10-20%	10.92-21.96	MRSA in moderate-burden settings, ESBL-E in Eastern EU
High	high (>20-50%)	20-50%	21.96-55.06	MRSA in Southern/Eastern Europe (Greece, Romania), ESBL-E in high-burden settings

beta was calibrated via **grid search** (beta = 0.005-0.120, step 0.005; 24 values) combined with three community carriage scenarios (pi = 0.001, 0.010, 0.050), run across 10 parallel cluster jobs (3 replicates per job = 30 total replicates per beta x pi combination). The best-fit beta per tier was defined as the replicate whose achieved incidence was closest to the tier midpoint.

Seeding simulation

Seeding rules - eight candidate introduction hospitals were evaluated:

Seeding rules. NC = Network-central. FTR = Facility-type random. MCO = Medecine Chirurgie Obstetrique (acute care). SSR = Soins de Suite et de Readaptation (rehabilitation). MCO/SSR = mixed facility. All centrality metrics computed on the directed transfer network using igraph (betweenness: directed, unnormalised).

Seed rule	Type	Selection criterion
Network hub (NC)	Network-central (NC)	Hospital ranking first on ALL five centrality metrics simultaneously: in-degree, out-degree, betweenness centrality, total bed count, and outgoing transfer volume. In this network, all five metrics identified the same facility (750100125, MCO/SSR, Ile-de-France, 1,713 beds), consistent with a scale-free network structure.
Random MCO (FTR)	Facility-type random (FTR)	One acute care hospital (MCO) drawn uniformly at random per simulation replicate.
Random SSR (FTR)	Facility-type random (FTR)	One rehabilitation hospital (SSR) drawn uniformly at random per simulation replicate.
Random MCO/SSR (FTR)	Facility-type random (FTR)	One mixed acute/rehabilitation facility (MCO/SSR) drawn uniformly at random per simulation replicate.

Simulation parameters:

Seeding simulation parameters. SS = steady state.

Parameter	Value	Description
N_SEED_INF	5	Index cases introduced on day 0 in the seed hospital
Tmax	732 days	Simulation duration (2 x 366 days, cycling 2024 PMSI data annually)
gamma (gamma)	1/387 per day	Daily biological decolonisation rate; mean carriage ~387 days (ESBL-E)
pi_vec (pi)	0	Community carriage on admission - novel pathogen assumption (no external reservoir)
N_REP	10 per job x 10 jobs = 100 total	Replicates per (seed rule x tier) combination
SS_WINDOW	30 days	Last N days used to compute steady-state (SS) incidence metrics
Beta draws	Truncated normal (mean=mean_beta, sd from 95% CI width)	Per-hospital beta drawn independently; propagates calibration uncertainty

Outcome metric definitions

All outcomes are computed per simulation replicate. Analyses use non-extinct replicates only unless stated otherwise.

Primary outcome metric definitions. I_loc = infected (colonised) patients per hospital. N_t = total occupied beds on day t. new_inf_t = new colonisations on day t. SS = steady state. pd = patient-days. HAI = healthcare-associated infection.

Metric	Abbreviation	Scale	Exact formula	Time window	Interpretation
Network steady-state incidence	net_ss_inc	National (bed-weighted)	$\text{mean}[\text{sum}(\text{new_inf_t})/\text{sum}(\text{N_t}) \times 1,000]$ over last SS_WINDOW days	Last 30 days	National hospital-acquired incidence rate per 1,000 patient-days at near-endemic equilibrium. Directly comparable to SPARES and ECDC HAI surveillance.
Facility-level mean steady-state incidence	fac_ss_inc_mean	Facility (unweighted)	mean over active hospitals of $[\text{new_inf_t}[h]/\text{N_t}[h] \times 1,000]$, then mean over last 30 days	Last 30 days	Mean of per-hospital daily acquisition rates, giving equal weight to each facility regardless of size. Closer to facility-level HAI reporting.
Network cumulative incidence	net_cumul_inc	National (bed-weighted)	$\text{mean}[\text{sum}(\text{new_inf_t})/\text{sum}(\text{N_t}) \times 1,000]$ over all Tmax days	All 732 days	Mean daily acquisition rate over the entire simulation. Reflects the integrated burden of colonisation throughout the epidemic trajectory, not just the end state.
Network steady-state prevalence	net_ss_prev_pct	National (bed-weighted)	$\text{mean}[\text{sum}(\text{I_loc_t})/\text{total_beds} \times 100]$ over last 30 days	Last 30 days	Proportion of total network beds occupied by colonised patients at near-endemic equilibrium. NOTE: values above 100% indicate a simulation capacity overflow bug and are capped at 100% in all displays.
Facility-level mean steady-state prevalence	fac_ss_prev_mean_pct	Facility (unweighted)	mean over hospitals of $[\text{I_loc_t}[h]/\text{beds}[h] \times 100]$, then mean over last 30 days	Last 30 days	Mean of per-hospital colonisation proportions. Unweighted: a 10-bed rural facility counts the same as a 1,000-bed university hospital.
Extinction	ever_extinct	•	$\text{sum}(\text{I_loc}) == 0$ at any point during the simulation	Any day 1-732	Pathogen died out - no colonised patients remain. With $\text{pi}=0$ (novel pathogen), extinction occurs when the pathogen fails to establish through transfers and within-hospital transmission alone.
Steady-state reached	ss_reached	•	$0.9 \leq \text{net_ss_inc} / \text{net_cumul_inc} \leq 1.1$	Ratio of last-30d mean to overall mean	The epidemic has stabilised: the last 30-day incidence is within 10% of the overall mean. Ratio > 1.1 = still growing; ratio < 0.9 = declining.

Results compilation

Local (PSOCK, Windows) and cluster (FORK, Linux) simulation results were compiled by `compile_seeding_novel.R`. Each source is loaded separately, tagged with a source label, and row-bound into a single `sim_summary` data frame. Each simulation replicate receives a globally unique identifier (`global_sim_id`) to prevent collision between local and cluster job indices. The compiled file saved to `Outputs/seeding_novel/compiled/` is the single input to this report.

Part 1 - Calibration results

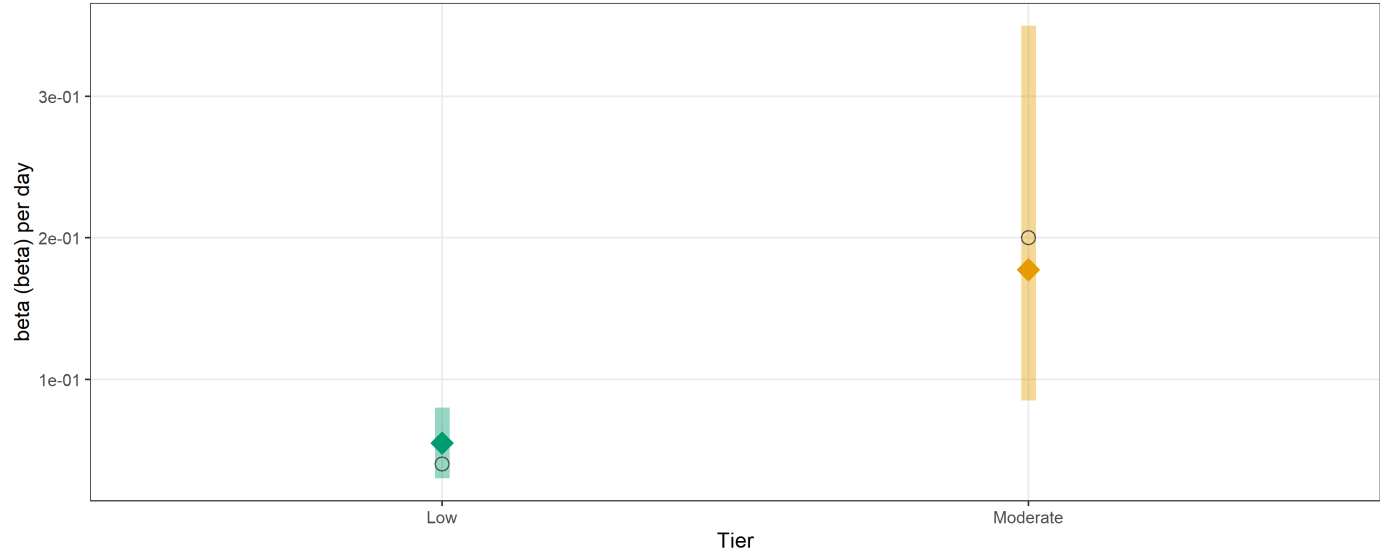
Beta grid search results. Incidence (inc) per 1,000 patient-days among non-extinct reps. SS = % of non-extinct reps reaching steady state. Prev = network prevalence on final day.

beta (beta)	N reps	Extinct (%)	SS (%)	Inc median /1k pd	Inc Q25	Inc Q75	Prev median
0.005	30	0	0.0	0.0058	0.0055	0.0059	0.0020
0.010	30	0	0.0	0.0165	0.0161	0.0168	0.0030
0.015	30	0	16.7	0.0475	0.0459	0.0482	0.0060
0.020	30	0	100.0	0.2092	0.2053	0.2198	0.0235
0.025	30	0	100.0	1.1921	1.1664	1.2284	0.1285
0.030	30	0	100.0	3.7179	3.6893	3.7633	0.3555
0.035	30	0	100.0	6.0723	6.0562	6.0837	0.5300

beta (beta)	N reps	Extinct (%)	SS (%)	Inc median /1k pd	Inc Q25	Inc Q75	Prev median
0.040	30	0	100.0	7.5785	7.5751	7.5837	0.6381
0.045	30	0	100.0	8.7478	8.7413	8.7532	0.7215
0.050	30	0	100.0	9.7445	9.7413	9.7496	0.7898
0.055	30	0	100.0	10.6283	10.6241	10.6334	0.8479
0.060	30	0	100.0	11.4184	11.4101	11.4255	0.8980
0.065	30	0	100.0	12.1374	12.1338	12.1438	0.9415
0.070	30	0	100.0	12.7999	12.7916	12.8023	0.9810
0.075	30	0	100.0	13.4086	13.3964	13.4144	1.0146
0.080	30	0	100.0	13.9770	13.9682	13.9803	1.0464
0.085	30	0	100.0	14.5161	14.5041	14.5209	1.0735
0.090	30	0	100.0	15.0129	15.0080	15.0225	1.0998
0.095	30	0	100.0	15.4884	15.4802	15.4953	1.1227
0.100	30	0	100.0	15.9400	15.9290	15.9419	1.1439
0.105	30	0	100.0	16.3636	16.3555	16.3710	1.1645
0.110	30	0	100.0	16.7754	16.7650	16.7791	1.1829
0.115	30	0	100.0	17.1606	17.1508	17.1697	1.2003
0.120	30	0	100.0	17.5328	17.5239	17.5392	1.2163
0.125	30	0	100.0	17.8921	17.8812	17.8958	1.2317
0.150	30	0	100.0	19.4781	19.4633	19.4894	1.2958
0.175	30	0	100.0	20.8075	20.8018	20.8145	1.3440
0.200	30	0	100.0	21.9592	21.9448	21.9654	1.3837
0.225	30	0	100.0	22.9725	22.9631	22.9824	1.4171
0.250	30	0	100.0	23.8895	23.8789	23.8993	1.4443
0.275	30	0	100.0	24.7470	24.7394	24.7547	1.4703
0.300	30	0	100.0	25.5814	25.5696	25.5934	1.4913
0.325	30	0	100.0	26.3484	26.3360	26.3702	1.5119
0.350	30	0	100.0	27.0398	27.0157	27.0511	1.5295

Calibrated beta (beta) per incidence tier

Diamond (*) = mean | Circle (o) = best-fit | Bar = 95% CI



Tier analysis: recommended beta values per ECDC incidence tier.

Tier	Inc low /1k pd	Inc high /1k pd	N reps	N jobs	Best beta	Mean beta	95% CI low	95% CI high	Mean inc achieved
Low	1.300	14.299	331	10	0.04	0.05491	0.030	0.08	9.9949
Moderate	14.299	28.742	540	10	0.20	0.17750	0.085	0.35	19.9713

Part 2 - Seeding simulation overview

Scenario: Novel pathogen | $\pi = 0$ (no community importation)

Sources: cluster_job_01, cluster_job_02, cluster_job_03, cluster_job_04, cluster_job_05, cluster_job_06, cluster_job_07, cluster_job_08, cluster_job_09, cluster_job_10

Total simulations: 1200 | **Tiers:** Low / Moderate | **Seed rules:** 4

Extinct: 68.4% | **Reached SS:** NaN% of non-extinct

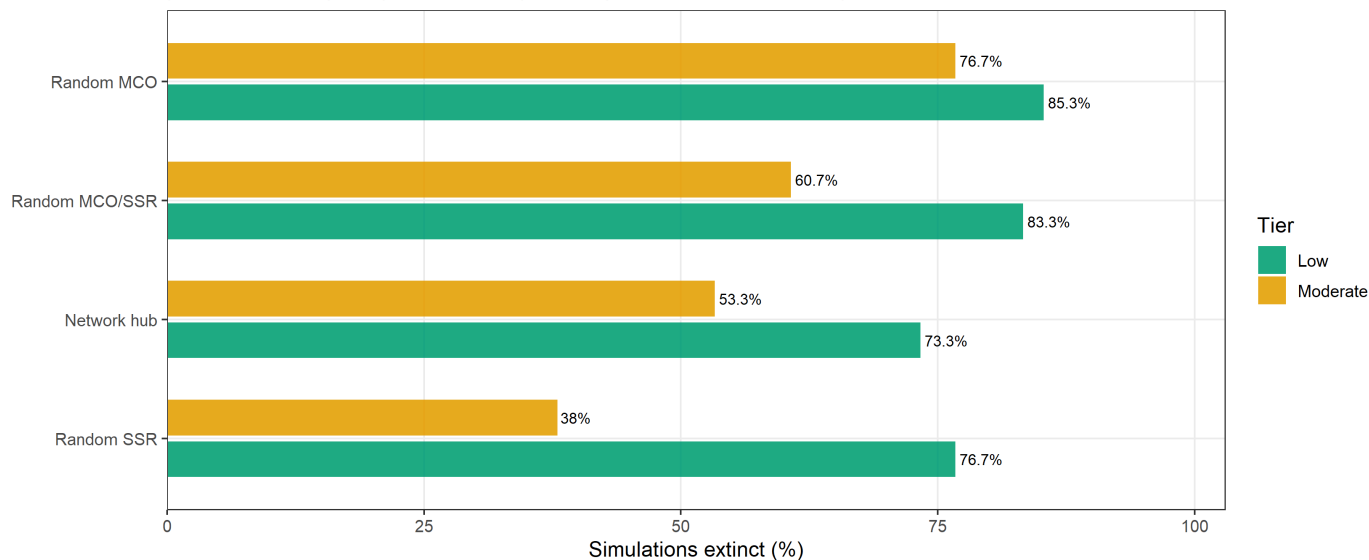
Key finding: 0 of 379 non-extinct sims still growing at day 732

Part 3 - Extinction and steady-state

3.1 Extinction probability by seed rule and tier

Extinction probability by seed rule and tier

$\pi = 0$: no community re-importation - ARB spreads only via transfers and within-hospital transmission



3.2 Steady-state indicator distribution

```
## > Steady-state indicator requires daily trajectories (traj_df). Load compiled RDS to enable.
```

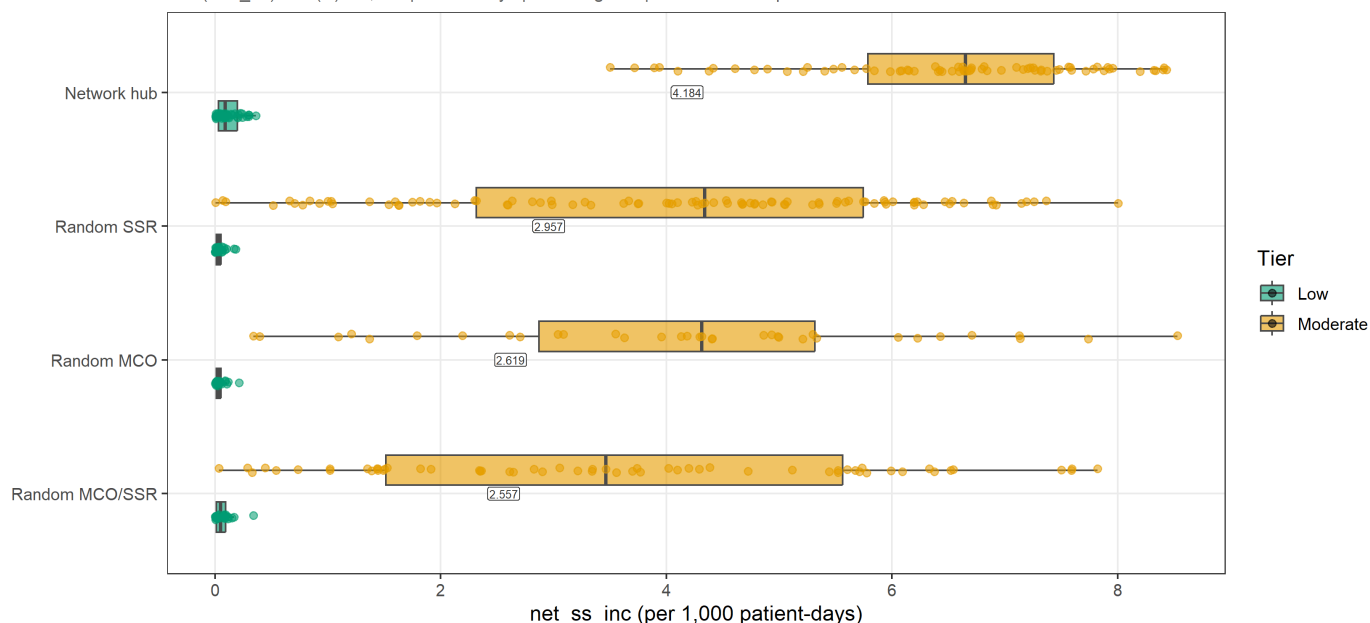
Part 4 - Primary outcome: Incidence per 1,000 patient-days

Box = IQR (25th-75th percentile). Whiskers = $1.5 \times \text{IQR}$. Each dot = one simulation replicate. Non-extinct replicates only.

4.1 National steady-state incidence (net_ss_inc) by seed rule

National (network-level) steady-state incidence (net_ss_inc) by seed rule

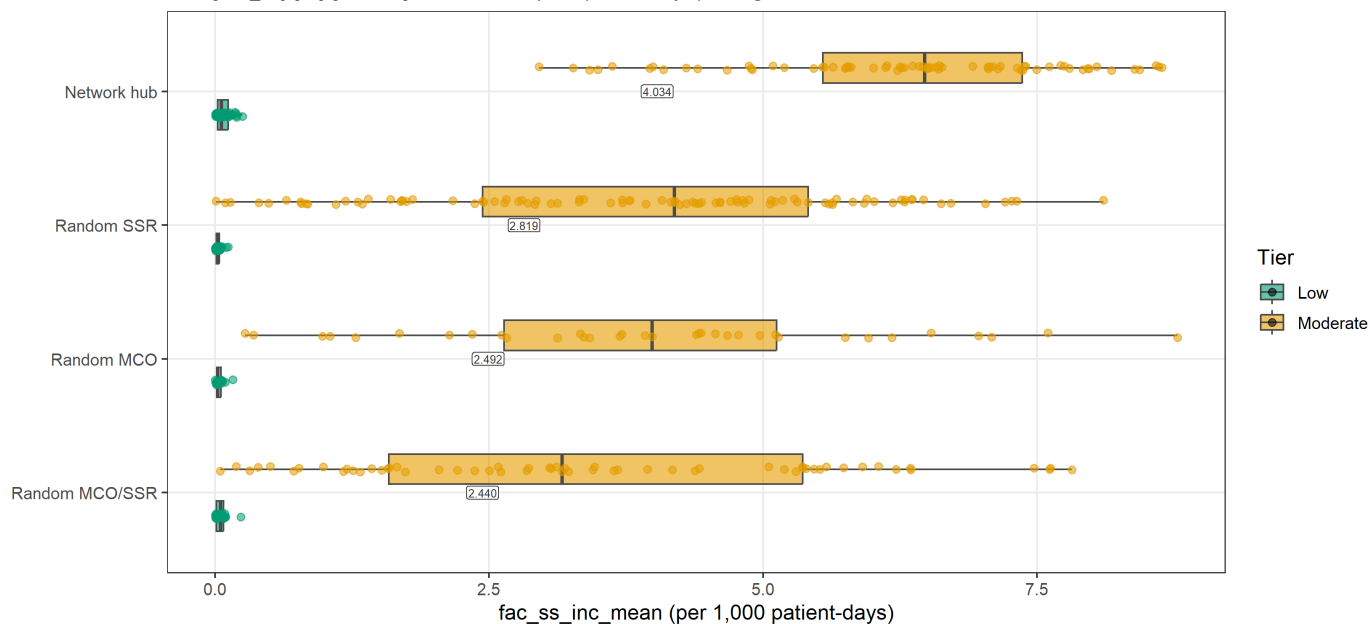
$\text{sum}(\text{new_inf})/\text{sum}(N) \times 1,000$ | last 30 days | bed-weighted | each dot = 1 replicate



4.2 Facility-level mean steady-state incidence (fac_ss_inc_mean) by seed rule

Facility-level mean steady-state incidence (fac_ss_inc_mean) by seed rule

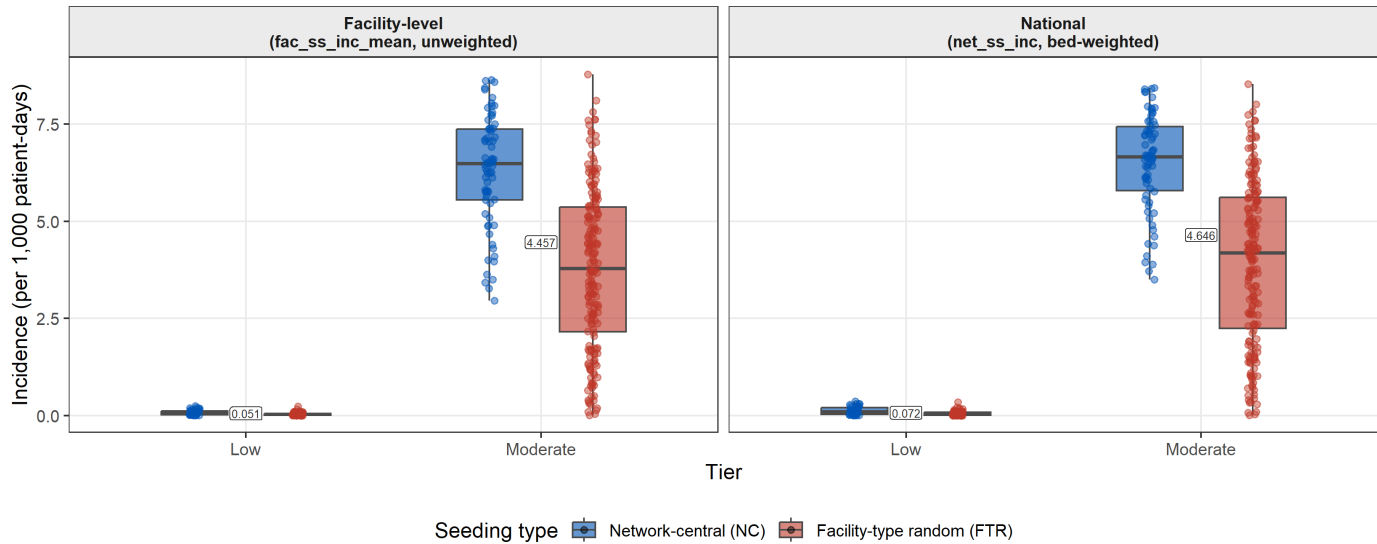
$\text{mean}[\text{new_inf}[h]/N[h] \times 1,000]$ over active hospitals | last 30 days | unweighted



4.3 National vs facility-level incidence - seeding type comparison

National (NC) vs facility-level (FTR) steady-state incidence by tier and seeding type

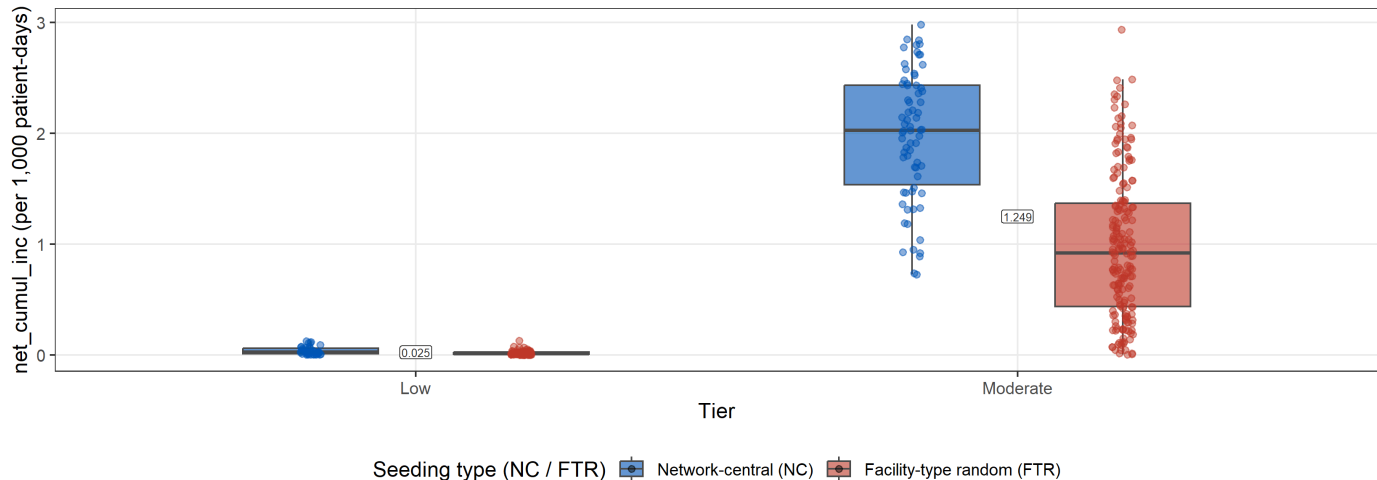
NC = Network-central | FTR = Facility-type random | each dot = 1 replicate



4.4 Cumulative incidence (net_cumul_inc) over 732 days

Cumulative incidence (net_cumul_inc) over 732-day simulation

Mean daily rate over ALL 732 days | per 1,000 patient-days | NC = Network-central | FTR = Facility-type random



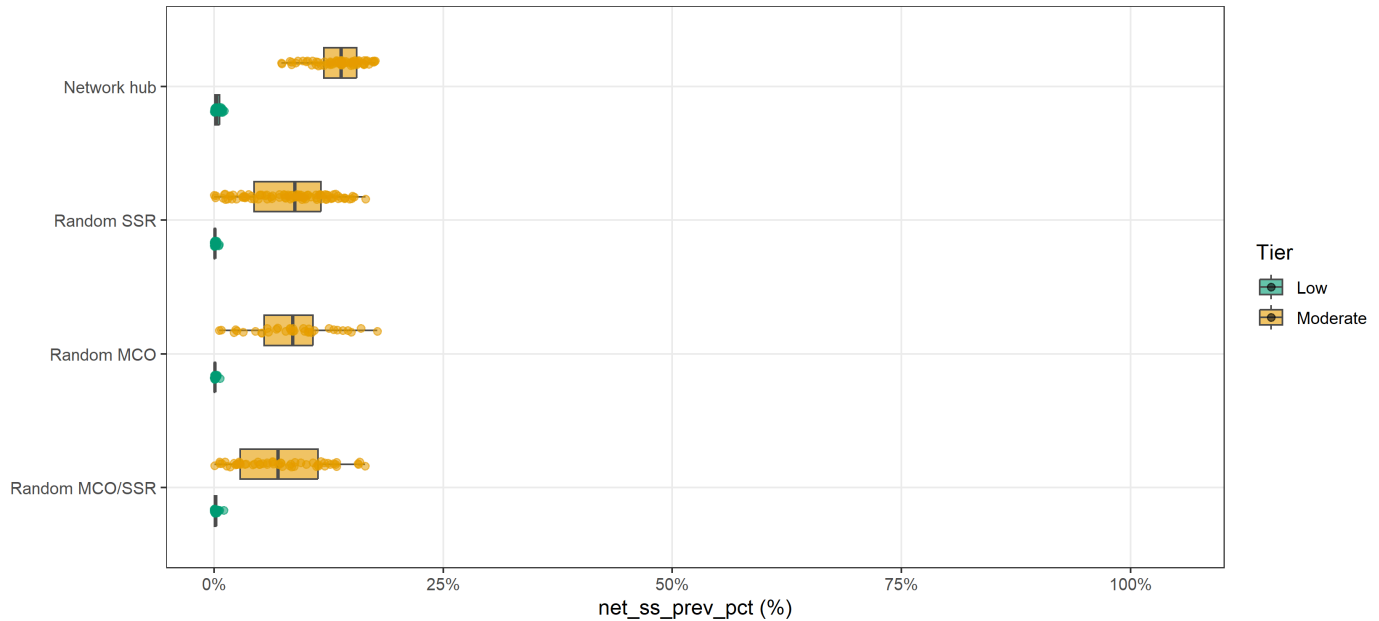
Part 5 - Prevalence (capped at 100%)

Note: Network-level prevalence (net_ss_prev_pct) and facility-level prevalence (fac_ss_prev_mean_pct) values above 100% are a **simulation capacity overflow artefact**: inter-hospital transfers can temporarily exceed a hospital's bed capacity when multiple source hospitals simultaneously route patients to the same destination. All prevalence values are capped at 100% in displays below. Incidence metrics (Part 4) are not affected.

5.1 National prevalence (net_ss_prev_pct, capped) by seed rule

National prevalence (net_ss_prev_pct, capped at 100%) by seed rule

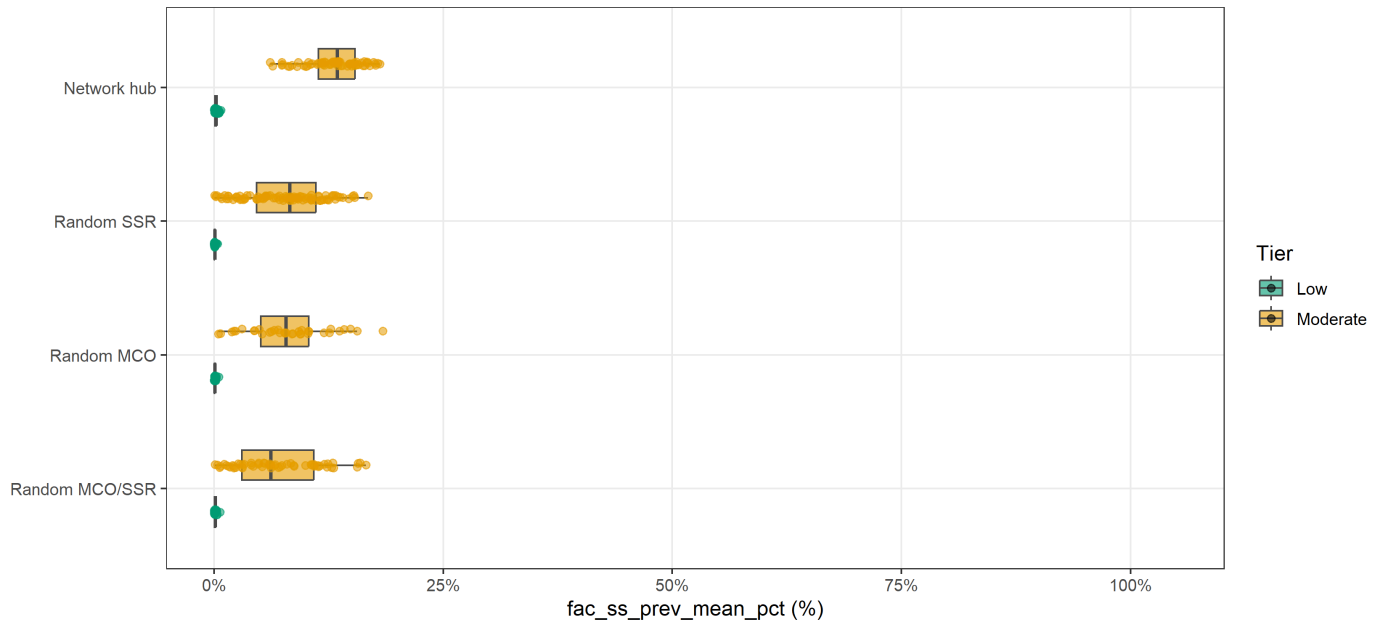
sum(l_loc)/total_beds x 100 | last 30 days | bed-weighted | capped at 100%



5.2 Facility-level prevalence (fac_ss_prev_mean_pct, capped) by seed rule

Facility-level prevalence (fac_ss_prev_mean_pct, capped at 100%) by seed rule

mean[l_loc[h]/beds[h]] x 100 | last 30 days | unweighted across hospitals



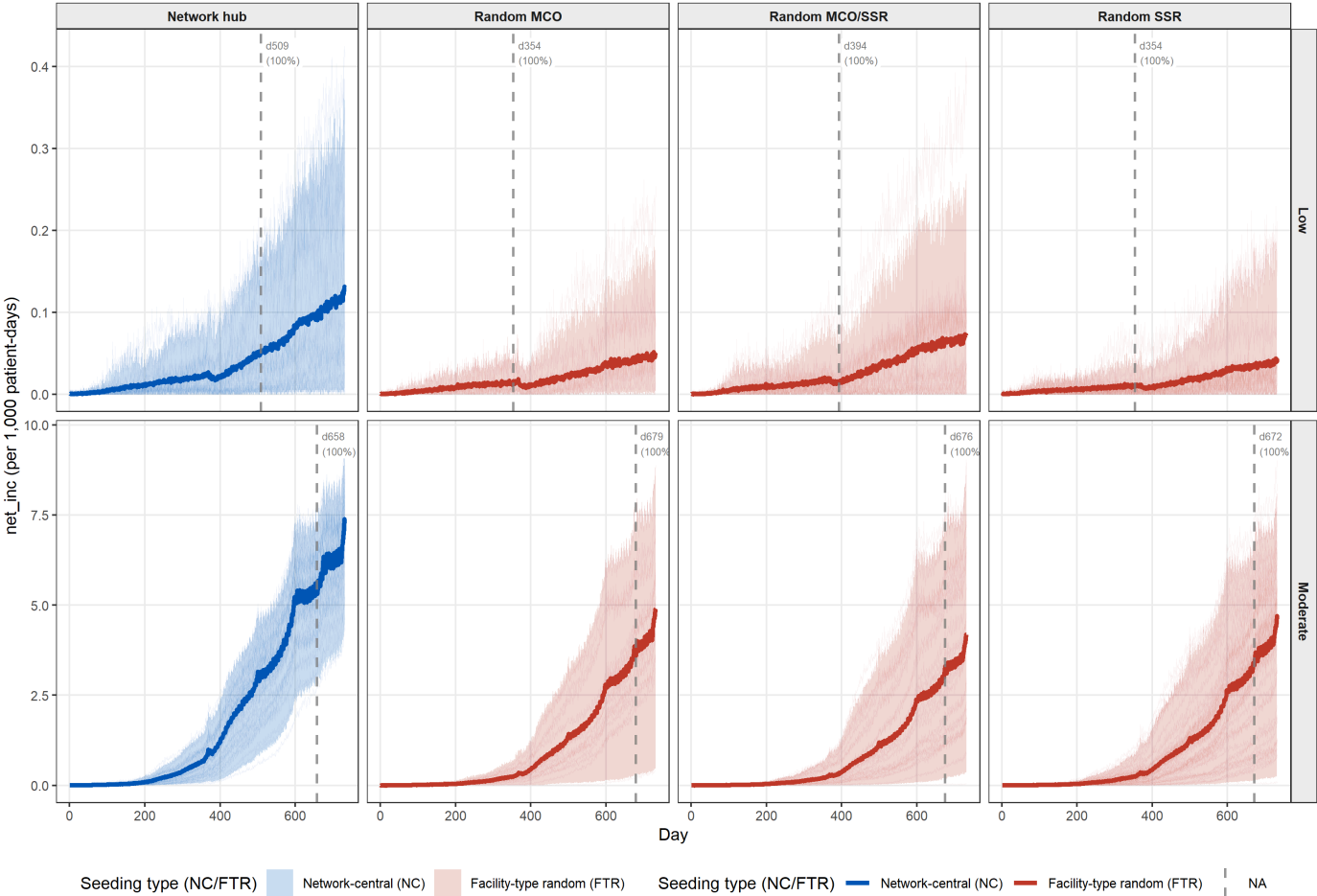
Part 6 - Epidemic trajectories

6.1 National incidence trajectory (net_inc)

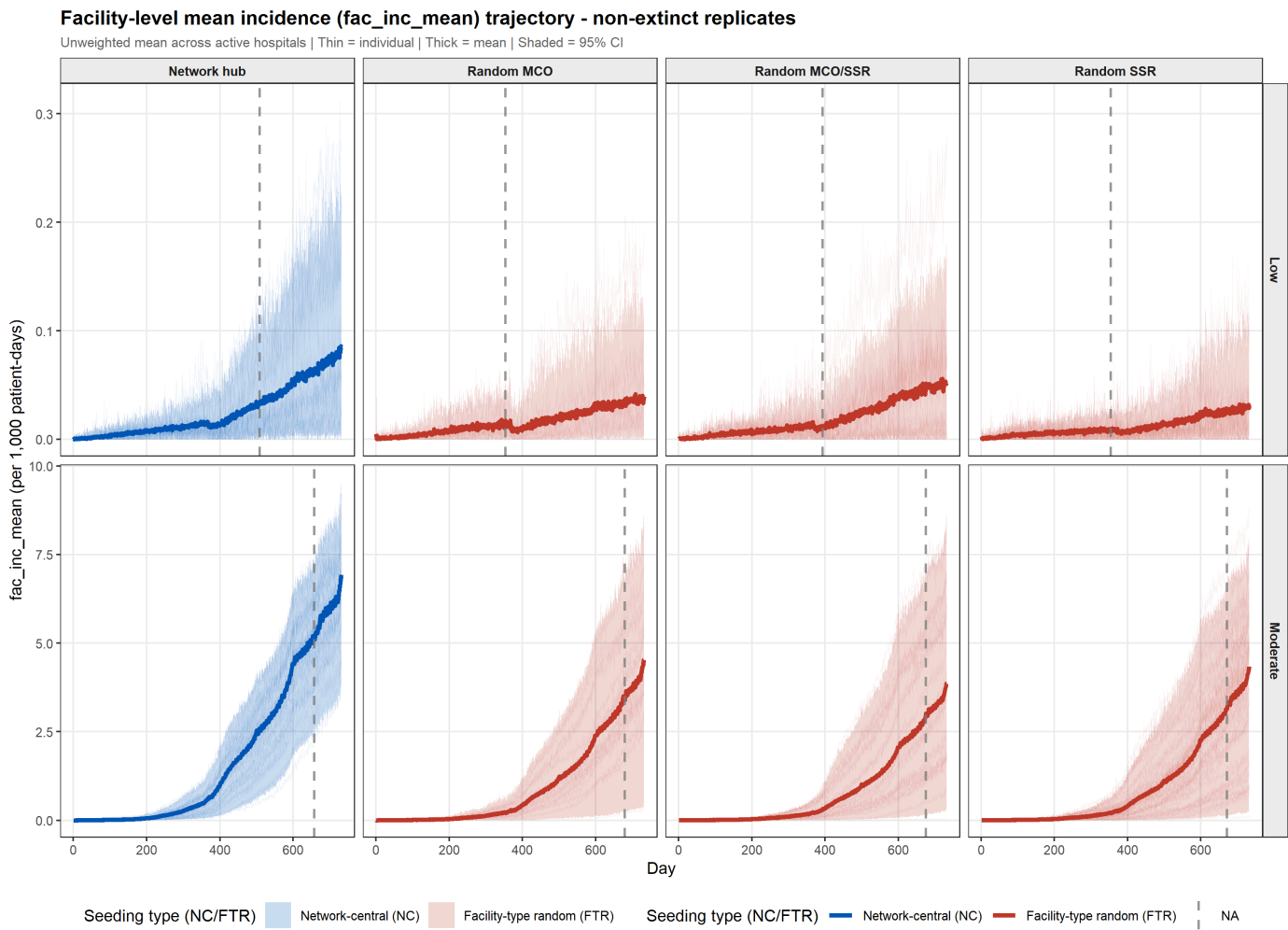
Thin lines = individual replicates. Thick line = mean. Shaded = 95% CI. Vertical dashed = mean day reaching SS (90% of final level).

National incidence (net_inc) trajectory - non-extinct replicates

Thin = individual sim | Thick = mean | Shaded = 95% CI | Dashed = mean day reaching 90% of SS level



6.2 Facility-level incidence trajectory (fac_inc_mean)



6.3 Steady-state day summary

Steady state (SS) day per simulation cell. SS = first day net_inc >= 90% of that replicate's net_ss_inc. NC = Network-central. FTR = Facility-type random. % reps reached SS = fraction of non-extinct replicates crossing the threshold before day 732.

Tier	Seeding type	Seed rule	Mean SS day	% reps reached SS
Low	NA	Network hub	509	100
		Random MCO	354	100
		Random MCO/SSR	394	100
		Random SSR	354	100
Moderate		Network hub	658	100
		Random MCO	679	100
		Random MCO/SSR	676	100
		Random SSR	672	100

Part 7 - Summary table

Summary table. Mean (SD) across non-extinct replicates. NC = Network-central. FTR = Facility-type random. net_ss_inc = national bed-weighted SS incidence. fac_ss_inc_mean = facility-level unweighted SS incidence. net_cumul_inc = mean daily national incidence over 732 days. net_ss_prev = national prevalence (capped at 100%). /1k pd = per 1,000 patient-days. SS = last 30 days.

Tier	Seeding type	Seed rule	N	Extinct (%)	SS reached (%)	net_ss_inc /1k pd	fac_ss_inc_mean /1k pd	net_cumul_inc /1k pd	net_ss_prev % (capped)
------	--------------	-----------	---	-------------	----------------	-------------------	------------------------	----------------------	------------------------

Tier	Seeding type	Seed rule	N	Extinct (%)	SS reached (%)	net_ss_inc /1k pd	fac_ss_inc_mean /1k pd	net_cumul_inc /1k pd	net_ss_prev % (capped)
Low	Network-central (NC)	Network hub	150	73.3%	8.7%	0.117 (0.101)	0.078 (0.065)	0.038 (0.035)	0.3 (0.3)
	Facility-type random (FTR)	Random MCO	150	85.3%	6%	0.046 (0.050)	0.037 (0.036)	0.018 (0.018)	0.1 (0.1)
		Random MCO/SSR	150	83.3%	6%	0.068 (0.075)	0.051 (0.051)	0.025 (0.028)	0.2 (0.2)
		Random SSR	150	76.7%	6%	0.039 (0.042)	0.029 (0.026)	0.015 (0.014)	0.1 (0.1)
Moderate	Network-central (NC)	Network hub	150	53.3%	7.3%	6.508 (1.260)	6.295 (1.434)	1.973 (0.581)	13.5 (2.6)
	Facility-type random (FTR)	Random MCO	150	76.7%	1.3%	4.236 (2.080)	4.036 (2.082)	1.053 (0.666)	8.5 (4.4)
		Random MCO/SSR	150	60.7%	0%	3.612 (2.174)	3.452 (2.149)	0.895 (0.661)	7.2 (4.6)
		Random SSR	150	38%	2.7%	4.056 (2.061)	3.869 (2.009)	1.003 (0.639)	8.1 (4.3)

Part 9 - Export

Saved to seeding_novel_report_table.csv

tier	seed_type	seed_rule	n_total	n_alive	pct_extinct	pct_ss_reached	mean_net_ss_inc	sd_net_ss_inc	mean_fac_ss_inc	sd_fac_ss_inc	mean_net_cumul_inc	sd_net_cumul_inc	mean_net_prev_capped	sd_net_prev_capped
Low	Network-central (NC)	Network hub	150	40	73.3	8.7	0.1171	0.1008	0.0775	0.0652	0.0380	0.0349	0.35	0.31
Low	Facility-type random (FTR)	Random MCO	150	22	85.3	6.0	0.0463	0.0501	0.0369	0.0360	0.0180	0.0177	0.13	0.14
Low	Facility-type random (FTR)	Random MCO/SSR	150	25	83.3	6.0	0.0680	0.0752	0.0506	0.0508	0.0255	0.0280	0.21	0.24
Low	Facility-type random (FTR)	Random SSR	150	35	76.7	6.0	0.0388	0.0424	0.0289	0.0256	0.0146	0.0144	0.11	0.13
Moderate	Network-central (NC)	Network hub	150	70	53.3	7.3	6.5077	1.2603	6.2948	1.4339	1.9733	0.5806	13.53	2.63
Moderate	Facility-type random (FTR)	Random MCO	150	35	76.7	1.3	4.2358	2.0801	4.0357	2.0823	1.0534	0.6664	8.53	4.41
Moderate	Facility-type random (FTR)	Random MCO/SSR	150	59	60.7	0.0	3.6118	2.1741	3.4524	2.1489	0.8953	0.6613	7.24	4.59
Moderate	Facility-type random (FTR)	Random SSR	150	93	38.0	2.7	4.0557	2.0613	3.8691	2.0088	1.0031	0.6389	8.14	4.35

2026-09-07 16:05 | Sims: 1200 | Non-extinct: 379 (31.6%) | SS: 57 of 379 | Traj: 878,400